Jihwan Lee

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Education

University of California, Irvine (GPA: 3.78) **Expected June 2028**
Bachelor of Science in Computer Science *Irvine, California*

- **Relevant Coursework:** Data Structures, Machine Learning, Operating Systems and Compilers, Probability/Statistics, Database Management, Embedded Systems, Computer Architecture, Graph Algorithms

Experience

University of California, Irvine **March. 2026 - Present**
Undergraduate Researcher *Irvine, California*

- Utilized PyTorch and MACE in Python to model molecular potential energy landscapes for protein-ligand interactions.
- Integrated custom force based algorithms with OpenMM and PLUMED to simulate and benchmark molecular dynamics.
- Containerized simulation environments with Docker to support reproducible experiments.

University of California, Irvine **Nov. 2025 – Feb. 2026**
ICS 33 - Intermediate Programming Undergraduate Learning Assistant *Irvine, California*

- Supported TAs in labs by assisting students with assignments and projects on advanced Python mechanics.
- Assisted in weekly lectures of 200+ students, providing individualized support for the students.
- Facilitated learning online by monitoring and answering students' questions on class discussion boards.

Projects

Watchdog | *Python, FastAPI, Gemini API, OpenCV, SQLite, WebSockets*
https://devpost.com/software/watchdog-yjrmkv

- Policy driven video monitoring for safety with AI assisted alerts. Submitted for LA Hacks AI Hackathon 2026.
- Integrated asynchronous Gemini verification with bounded queues, rate limiting, and cached results.
- Developed FastAPI endpoints, WebSocket alerts, SQLite storage, and automated image/video evidence capture.

Tennis Stroke Comparison | *Python, MediaPipe, OpenCV, JavaScript, HTML5 Canvas*
https://jihwantime.github.io/tennis-stroke-comparison/

- Built a Python CLI that extracts 33-keypoint pose skeletons from tennis video using MediaPipe Pose Landmarker.
- Engineered frame-accurate skeleton overlays on raw video using HTML5 Canvas and requestVideoFrameCallback.

GameSetMatch | *TypeScript, React, Postgres, Next.js, Python, Scikit-Learn*
https://gamesetmatch.vercel.app/

- Built a tennis match history web app with TypeScript using Postgres, with a Next.js and React frontend.
- Created a Python ETL pipeline ingesting 80K+ matches from a public archive into Postgres database.
- Trained a Scikit-Learn performance-rating model and a surface-aware Elo match predictor validated at 70% accuracy.

SQL Airport Database Project | *Python, SQL*
http://learning-to-fly.onrender.com/

- Developed an event-based Python backend engine to bridge tkinter GUI with an SQLite database.
- Implemented secure SQL queries to perform CRUD operations across multiple relational database tables.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, HTML/CSS, Cuda, SQL
Technologies: Git, Docker, React, Tailwind, Next.js, PostgreSQL, Scikit-Learn, NumPy, Matploblib, Pandas, PyTorch, OpenCV
Interests: Tennis, Hiking, Music, League of Legends

Leadership / Extracurricular

Tennis Club **Spring 2025 – Present**
Vice President *University of California, Irvine*

- Part of the Executive Board. Main job is to supervise and provide guidance for the Competitive Team.
- Address concerns brought up between competitive board members or competitive team members, and executive board.
- Work with fundraiser chair to raise 75% of our annual Sport Club allocation funds.